MODULE 2

The Math Toolkit

This module teaches every piece of math used later in the course. If you already know it, skim the worked examples and move on.

2.1 Big and small numbers (scientific notation)

SIMPLE APPROACH

Light is *so* fast that it could go around the Earth about seven times in one second. To write such big numbers — and tiny ones like "one billionth of a second" — scientists use a shortcut: count the zeros and write that count as a little number up high. 10^3 means 1 followed by 3 zeros = 1000. 10^{-3} means "1 divided into 1000 pieces" = 0.001.

KEY EQUATION — Scientific notation

$$N = m \times 10^k \qquad \text{and} \qquad 10^a \times 10^b = 10^{a+b}$$

What every symbol means

Symbol	Meaning	Units
N	the number being written	any
m	the *mantissa*, a number between 1 and 10	—
k	the power of ten (how many places to move the decimal point; negative = move left)	—
a, b	any two exponents; multiplying powers of ten adds the exponents	—

Prefix	Symbol	Factor	Light travels about…
milli	m	10^{-3}	300 km in 1 ms
micro	μ	10^{-6}	300 m in 1 μs
nano	n	10^{-9}	30 cm in 1 ns
pico	p	10^{-12}	0.3 mm in 1 ps

WORKED EXAMPLE — multiplying with powers of ten

Given: speed $c = 2.997 \times 10^8$ m/s, time $t = 4.00 \times 10^{-7}$ s. Find $c \times t$.

Step 1 — multiply the mantissas

$2.997 \times 4.00 = 11.988$

Step 2 — add the exponents

$10^8 \times 10^{-7} = 10^{8+(-7)} = 10^1$

Step 3 — combine

$11.988 \times 10^1 = 119.88$ m

Result: $c\,t = 119.88$ m.

2.2 Angles: degrees and radians

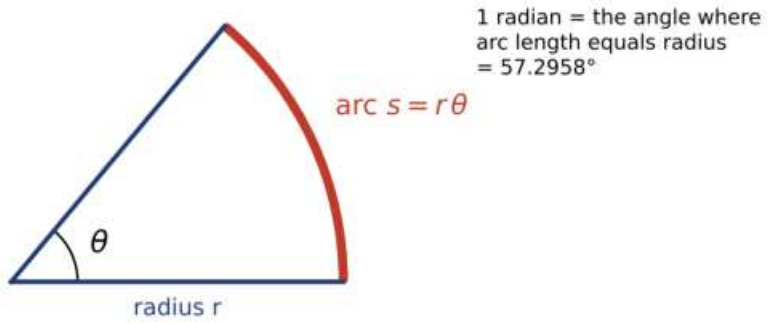


Figure 2.1: A radian is the angle whose arc is exactly one radius long.

SIMPLE APPROACH

Imagine a pizza. If you cut a slice whose crust is exactly as long as the line from the center to the crust, that slice's angle is **one radian**. A whole pizza has a bit more than 6 of those slices (6.28, which is 2π).

KEY EQUATION — Degree–radian conversion and arc length

$$\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180} \qquad s = r\,\theta_{\text{rad}}$$

What every symbol means

Symbol	Meaning	Units
θ_{deg}	an angle in degrees (a full circle = 360°)	degrees
θ_{rad}	the same angle in radians (a full circle = 2π)	radians
π	3.14159…, ratio of a circle's circumference to its diameter	—
r	radius (in LiDAR: the range to the target)	m
s	arc length: how far you move along the circle	m

Why care? Because **sideways error = range × angle error (in radians)**. This "small-angle rule" will appear in nearly every module. It works because for small angles the arc and the straight sideways distance are almost identical ($\sin\theta \approx \tan\theta \approx \theta$ for θ in radians).

WORKED EXAMPLE — a tiny angle at a long range

Given: an aiming error of 20 arc-seconds at 100 m.

Step 1 — seconds to degrees

$20'' = 20/3600 = 0.005556°$

Step 2 — degrees to radians

$0.005556 \times \pi/180 = 9.70 \times 10^{-5}$ rad

Step 3 — arc length
$s = 100 \times 9.70 \times 10^{-5} = 0.0097$ m

Result: about 1 cm sideways — the classic surveyor's rule "20 seconds at 100 m is a centimeter."

2.3 Sine and cosine

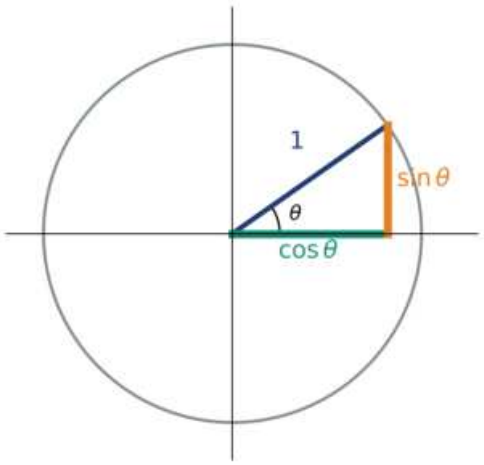


Figure 2.2: On a circle of radius 1, cosine is the "across" part and sine is the "up" part.

SIMPLE APPROACH

Lean a 1-meter stick against a wall. The bottom of the stick is some distance from the wall (that is **cosine**) and the top is some height up the wall (that is **sine**). If the stick leans a lot, cosine is big and sine is small. If it stands almost straight up, sine is big and cosine is small.

KEY EQUATION — Right-triangle trigonometry

$$\text{across} = L\cos\theta, \qquad \text{up} = L\sin\theta, \qquad L = \sqrt{\text{across}^2 + \text{up}^2}, \qquad \theta = \text{atan2}(\text{up}, \text{across})$$

What every symbol means

Symbol	Meaning	Units
L	length of the slanted side (hypotenuse)	m
θ	angle measured from the "across" direction	deg or rad
cos, sin	functions that turn an angle into a number between -1 and $+1$	—
atan2	"two-argument arctangent": returns the angle and gets the quadrant right	rad

WORKED EXAMPLE — a 3-4-5 triangle

Given: a slope distance of 5.000 m at an angle of 36.87° above horizontal.

Step 1
$\text{across} = 5.000\cos 36.87^\circ = 5.000 \times 0.8000 = 4.000$ m

Step 2

$\text{up} = 5.000 \sin 36.87^\circ = 5.000 \times 0.6000 = 3.000$ m

Step 3 (check)

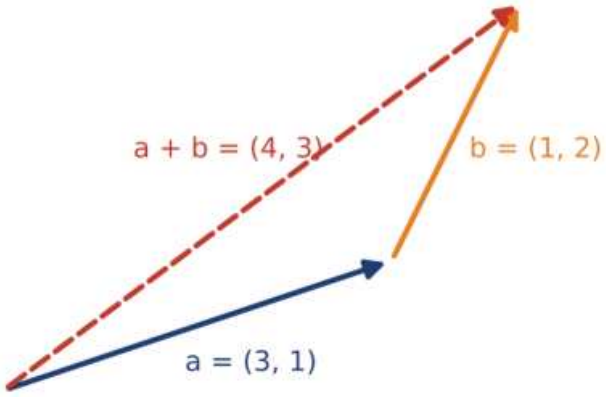
$\sqrt{4^2 + 3^2} = \sqrt{25} = 5$ ✓

Result: 4.000 m across and 3.000 m up.

2.4 Vectors

SIMPLE APPROACH

A vector is an arrow: it says *how far* and *which way*. "Walk 3 steps east and 1 step north" is a vector. Two walks one after the other add up to one bigger walk.



ENGINEER'S VIEW

A 3-D vector is written $\mathbf{v} = (v_x, v_y, v_z)^\mathsf{T}$. Vectors add component by component. Its length (norm) is $\|\mathbf{v}\| = \sqrt{v_x^2 + v_y^2 + v_z^2}$. Two products appear later:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \quad \text{(dot: a number)}, \qquad \mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{pmatrix} \quad \text{(cross: a vector)}.$$

The dot product measures how much two arrows point the same way. The cross product is perpendicular to both, and its length equals the area of the parallelogram they span — which is why LIO-SAM uses it to measure distances to lines and planes (Module 11).

WORKED EXAMPLE — dot and cross products

Given: $\mathbf{a} = (1, 2, 0)$, $\mathbf{b} = (3, 0, 0)$.

Dot

$1 \cdot 3 + 2 \cdot 0 + 0 \cdot 0 = 3$

Cross

$(2 \cdot 0 - 0 \cdot 0,\ 0 \cdot 3 - 1 \cdot 0,\ 1 \cdot 0 - 2 \cdot 3) = (0, 0, -6)$

Meaning

The parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$ has area $\|(0, 0, -6)\| = 6$; with base $\|\mathbf{b}\| = 3$ its height is $6/3 = 2$ — exactly the distance from the tip of $\mathbf{a}$ to the line along $\mathbf{b}$.

Result: dot = 3, cross = (0, 0, −6), point-to-line distance = 2.

2.5 Matrices in one page

ENGINEER'S VIEW

A matrix is a table of numbers that transforms a vector. Multiplying a 3×3 matrix by a vector gives a new vector whose each entry is a row of the matrix "dotted" with the vector:

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ax + by + cz \\ dx + ey + fz \\ gx + hy + iz \end{pmatrix}.$$

Key facts used later: the **transpose** $\mathbf{A}^{\mathsf{T}}$ swaps rows and columns; the **identity** $\mathbf{I}$ changes nothing; the **inverse** $\mathbf{A}^{-1}$ undoes $\mathbf{A}$; for a *rotation* matrix the inverse is simply the transpose, $\mathbf{R}^{-1} = \mathbf{R}^{\mathsf{T}}$.

WORKED EXAMPLE — matrix times vector

$$\begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2\cdot 1 + 0\cdot 4 + 1\cdot 2 \\ 0\cdot 1 + 1\cdot 4 + 0\cdot 2 \\ 1\cdot 1 + 0\cdot 4 + 3\cdot 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 7 \end{pmatrix}$$

2.6 Statistics: mean, standard deviation, RMSE

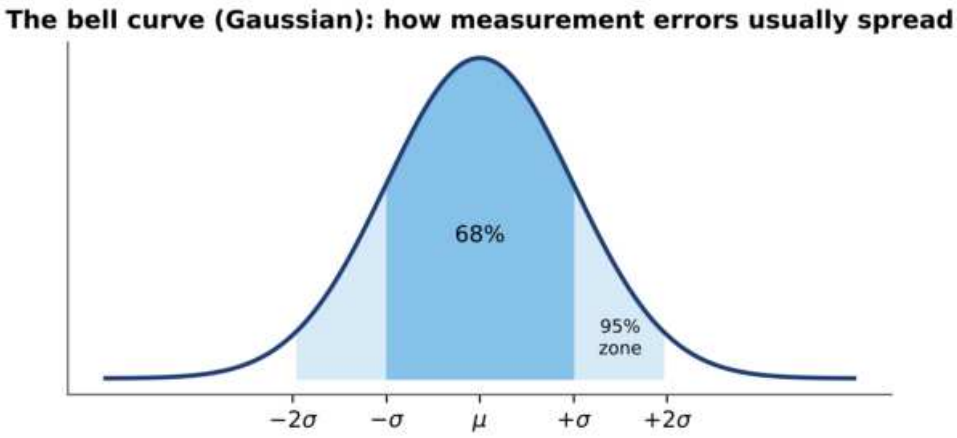


Figure 2.3: Random measurement errors usually follow a bell curve. About 68% fall within one standard deviation σ and 95% within 1.96σ.

SIMPLE APPROACH

Throw ten paper balls at a target. The **mean** tells you where they landed on average (are you aiming too far left?). The **spread** (standard deviation) tells you how scattered they are. A great thrower has a mean right on the bullseye *and* a small spread.

KEY EQUATION — Mean, standard deviation, RMSE

$$\bar{e} = \frac{1}{n}\sum_{i=1}^{n} e_i, \qquad s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(e_i - \bar{e})^2}, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}$$

What every symbol means

Symbol	Meaning	Units
e_i	the i-th error (measured minus true)	m
n	how many errors we have	—
$\sum$	"add them all up," from $i = 1$ to n	—
$\bar{e}$	mean error = *bias* (systematic part)	m
s	sample standard deviation = *precision* (random part); $n-1$ because one degree of freedom was spent computing $\bar{e}$	m
RMSE	root-mean-square error = total *accuracy*; combines bias and spread: $\text{RMSE}^2 \approx \bar{e}^2 + s^2$	m

WORKED EXAMPLE — five elevation errors

Given: errors $+2, -1, +3, 0, +1$ cm.

Mean

$(2 - 1 + 3 + 0 + 1)/5 = 5/5 = 1.0$ cm (a 1 cm high bias)

Deviations from mean

$1, -2, 2, -1, 0$; squares $1, 4, 4, 1, 0$; sum $= 10$

Std. dev.

$s = \sqrt{10/4} = \sqrt{2.5} = 1.58$ cm

RMSE

squares of errors $4, 1, 9, 0, 1$; sum $= 15$; $\text{RMSE} = \sqrt{15/5} = \sqrt{3} = 1.73$ cm

Check

$\bar{e}^2 + s^2 \cdot \frac{n-1}{n} = 1 + 2.5 \times 0.8 = 3.0 = \text{RMSE}^2$ ✓

Result: bias 1.0 cm, precision 1.58 cm, accuracy (RMSE) 1.73 cm.

2.7 Adding independent errors: root-sum-square

KEY EQUATION — Propagation of independent errors

$$\sigma_{\text{total}} = \sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2}$$

What every symbol means

Symbol	Meaning	Units
σ_j	standard deviation of the j-th independent error source	m
σ_{total}	standard deviation of their sum	m

Errors do *not* add like ordinary numbers because random errors partly cancel. Two independent 3 cm and 4 cm errors combine to $\sqrt{9+16} = 5$ cm, not 7 cm. The rule only holds when the sources

are *independent*; a systematic error that pushes every point the same way must be treated separately (see Module 13).

TRY IT YOURSELF

2.1 Convert $0.05°$ to radians, then find the sideways error it causes at 80 m.
2.2 Compute $\sqrt{2^2 + 1.5^2 + 3^2}$.
2.3 A wall is 12 m across and 5 m up from you along the slope. How long is the slope distance?
2.4 (Simple) You need 3 big steps forward and then 4 big steps to the right. If you could walk straight there instead, how many steps would it be?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 158.

SOURCES FOR THIS MODULE

Ghilani, C. D. (2017), *Adjustment Computations: Spatial Data Analysis*, 6th ed., Wiley (statistics, RMSE, propagation); NIST SP 811, *Guide for the Use of the International System of Units*.